

LoadLead — UI/UX & Defect Audit Report

Date: 2026-07-04 · **Discipline:** UX Research · UI Design · QA Automation · **Type:** Non-destructive audit (read-only). No code, layout, or live-environment changes were made. **Approval required before any implementation.**

Surface audited: `frontend-v2` (React + Vite SPA) — all five personas: Shipper, Driver, Owner-Operator, Carrier (dispatcher/admin), Platform Admin, plus Receiver. Routes mapped from `src/App.tsx` ; hierarchy and defects traced to component source (`file:line` cited throughout).

1. Executive Summary

The platform is **feature-complete and internally consistent** — a shared design system (`components/ui` , `PageHeader` , `atoms.tsx`), consistent auth-gated routing (`RequireRole`), loading states across 88 call sites, an `ErrorBoundary` , and alt text on all 10 images. The **Driver dashboard is the model to copy**: it surfaces the single time-critical thing (live offers + a live `Countdown` to accept/decline) above the fold.

The core issue is **hierarchy and flow, not missing features**. Several dashboards present information as a **flat list or a set of equal-weight tabs**, so the user's most urgent action is not "above the fold." The three highest-impact patterns:

1. **The Carrier loadboard ("Dispatch") is hidden behind a tab** — a dispatcher's primary job is one click removed from their landing screen.
2. **The Shipper dashboard leads with a *hardcoded, fake analytics* chart** and a flat load list — no "needs-attention" grouping, and a fabricated "Match velocity" graph that misrepresents data.
3. **Settings is a 1,849-line, 7-tab monolith** where a **critical onboarding gate (Verification) is split across two separate tabs**, buried two levels deep.

Critical blockers discovered

- **None that block usage.** No broken routes (all `App.tsx` paths resolve to a component; `*` → `NotFound`). No missing auth guards. This is a **polish + information-architecture** engagement, not a firefight.
- **One data-integrity defect (High):** the Shipper "Match velocity (7 days)" chart renders **static fabricated values** (`ShipperDashboard.tsx:147`), presenting fake data as a real business metric.

Tally: 1 High, 5 Medium, 6 Low defects · 4 high-click / cognitive-load flags · full P1/P2/P3 re-org proposed for 3 dashboards + the loadboard. **Nothing is removed** — every re-org below relocates or regroups existing elements only.

2. UI Defect Log (Bug Tracking)

#	Severity	Component / Page	Defect	Suggested Fix (non-destructive)
D1	High	Shipper Dashboard (<code>pages/shipper/ShipperDashboard.tsx:144–152</code>)	"Match velocity (7 days)" chart renders hardcoded bar heights <code>[40, 55, 38, 70, 62, 88, 75]</code> with static day labels — fabricated data shown as a real metric.	Wire to a real match-rate endpoint, or relabel "Sample" / gate behind real data. Do not ship fake analytics. Keep the widget; feed it truth.

#	Severity	Component / Page	Defect	Suggested Fix (non-destructive)
D2	Med	Carrier Dashboard (<code>pages/carrier/CarrierDashboard.tsx:682–696</code>)	The loadboard is the "Dispatch" tab , 4th of 4 tabs — the dispatcher's core task is not the default view.	Make <code>dispatch</code> the default tab for dispatcher role, or promote an "Active loads" strip above the tab bar (see §4).
D3	Med	Settings (<code>pages/settings/SettingsPage.tsx:537–538, 822–823</code>)	Verification is split across two tabs ("ID Verification" + "Business Verification") inside a 7-tab page — a required onboarding gate is fragmented and buried.	Group into one "Verification" tab with two sub-sections; surface a completion badge on the tab. No fields removed.
D4	Med	Shipper "Post a load" CTA (<code>ShipperDashboard.tsx:60</code>)	CTA is <code>disabled={!profileComplete}</code> with no inline reason — a new shipper sees a dead button and no next step.	Keep disabled, add a tooltip/inline hint ("Complete your profile to post") linking to the exact settings tab.
D5	Med	Data tables — horizontal scroll (<code>ShipperDashboard, CarrierMembers, admin/OrgManagementPanel, admin/ComplianceConsole, admin/StaffManagement, admin/BetaProgramDashboard, carrier/CarrierDashboard</code>)	Wide tables use <code>overflow-x-auto + fixed min-w-[...]</code> , so on ≤1080p / tablet the table scrolls horizontally and key columns fall off-screen (the classic "grid breaks on 1080p" class).	Priority columns pinned; secondary columns collapse into a row-expander or stack on <lg. Responsive column strategy, not raw horizontal scroll.
D6	Med	Whole app — initial load (<code>dist/assets/index-*.js</code> ≈ 963 KB / 265 KB gzip , single chunk)	No code-splitting: every persona downloads all five dashboards' code on first paint → slow "slow-loading component" symptom, worst on drivers' mobile/poor connections.	Route-level <code>React.lazy()</code> + <code>manualChunks</code> for vendor (maps, charts). Ties to the "unresponsive/slow" scan item.
D7	Low	Settings (<code>pages/settings/SettingsPage.tsx</code> , 1,849 LOC / 7 tabs)	Single monolithic page mixing profile, equipment, authority, 2x verification, organisation, security → high scroll + tab-hunt cognitive load.	Split into logical groups (Identity, Business, Security, Org) — regroup, don't delete.
D8	Low	Owner-Operator Dashboard (<code>OwnerOperatorDashboard.tsx:191, 199</code>)	Fleet management actions deep-link to <code>settings?tab=fleet</code> ; the dashboard shows fleet read-only, so "manage" = leave dashboard → settings → tab.	Inline the common fleet action (add/assign driver) on the dashboard card; keep the full manager in settings.
D9	Low	Shipper Dashboard	No above-the-fold KPI/summary row and no urgent grouping — OPEN/OFFERED loads needing attention are visually equal to DELIVERED/CANCELLED in one flat list.	Add a StatCard summary row + "Needs attention" section (see §4). Uses existing <code>StatCard</code> .
D10	Low	Global	The only two <code>TOD0/placeholder</code> markers in	Grep-gate in CI; verify none render to users.

#	Severity	Component / Page	Defect	Suggested Fix (non-destructive)
			pages + components are benign, but confirm no persona sees literal placeholder copy in prod.	
D11	Low	Icon-only controls	Several <code><Button variant="ghost" size="sm"></code> are icon-only (e.g. row actions) — verify each has an <code>aria-label</code> / title for screen readers and 44px tap target on mobile.	Add <code>aria-label</code> + ensure min tap size.
D12	Low	Broken-link coverage	Static scan shows all <code><Link to></code> /route targets resolve; runtime dead-link/console-error coverage is not yet automated.	Add a Playwright crawl asserting 0 console errors + no 404 on nav across personas (QA automation deliverable).

Scan method: route map from `App.tsx` ; per-page render-hierarchy extraction; defect greps for `overflow-x` , fixed `min-w` , `TOD0` , alt/aria, tab/accordion density; bundle size from `vite build` . All findings are code-verified, not speculative.

3. Click-Path & Friction Analysis

High-click zones (>3 clicks / task, or core task not on landing)

Task (persona)	Current path	Clicks	Friction
Dispatch a load to a driver (Carrier)	Login → land on Overview tab → click Dispatch tab → locate load → open "pick a driver" select (<code>CarrierDashboard.tsx:551</code>) → choose → confirm	5+	The dispatcher's #1 job is behind a tab that isn't the default.
Complete verification (Carrier/OO onboarding gate)	Login → Settings (nav) → ID Verification tab → (later) Business Verification tab	3–4, split	A single gated goal is fragmented across two tabs of seven (D3).
Post first load (new Shipper)	Dashboard → "Post a load" disabled (D4) → hunt Settings → complete profile → back → Post a load → long form (<code>PostLoad.tsx</code> , 811 LOC)	6+	Dead CTA with no signposting; the unblock path is undiscoverable.
Manage fleet / assign driver (OO)	Dashboard → "Manage"/"Add driver" → navigates away to <code>settings?tab=fleet</code> → act → back to dashboard	3–4 + context switch	Read on one screen, act on another (D8).

Cognitive-load flags (too much unstructured data / buried info)

- **Settings (1,849 LOC, 7 tabs)** — the densest surface; verification, identity, equipment, org, and security all compete at the same level (D7).
- **Carrier Dashboard (701 LOC, 4 tabs)** and **Compliance Console (768)**, **Beta Program (730)** — dense tables with `min-w` horizontal scroll (D5) and no "needs-attention" summary.
- **Shipper Dashboard** — a **flat, undifferentiated load list** (all 7 statuses equal) topped by a **fake chart** (D1/D9): the eye is drawn to fabricated analytics instead of loads needing action.

What's already right (keep as the pattern)

- **Driver Dashboard** (`DriverDashboard.tsx:142–246`): title "Live load offers" → offers list → per-offer **Countdown** → Accept/Decline. The single time-critical action is first, unmissable, and time-boxed. **This is the P1 model** the other dashboards should imitate.

4. Proposed Information Hierarchy (Dashboards & Loadboards)

Principle: *surface the time-sensitive and the actionable; demote the historical and the analytical.* Every element below already exists — this is re-ordering and grouping only (Safety Constraint §honored).

4.1 Shipper Dashboard

Tier	Content (all existing)
P1 — first 2 seconds	Summary StatCard row (Open · Offered/awaiting · In-transit · Needs-action counts) + a "Needs attention" block: loads in OFFERED / DRAFT (action required). Primary Post a load CTA with the D4 hint.
P2 — 1 click	The full "All loads" list with its existing search + status filter (unchanged), default-sorted newest/active first.
P3 — scroll / sub-tab	"Match velocity" chart — only once wired to real data (D1) — plus completed/cancelled history.

Your loads

[Open 4]

[Awaiting 2]

[In-transit 1]

[Δ3]

NEEDS ATTENTION (OFFERED / DRAFT)

• Load #1042 OFFERED → Review offer

All loads

[search] [status ▼]

... list ...

► Match velocity (real data)

► History

P1

P2

P3

4.2 Carrier Dashboard (dispatcher)

Tier	Content (all existing)
P1	An "Active loads / Dispatch" strip above the tab bar (or <code>dispatch</code> as the default tab): open loads + the "pick a driver" assign control (<code>:551</code>). Urgent = unassigned/expiring first.
P2	The existing tabs — Overview, Drivers (onboard), Verification — remain one click away, order by frequency of use.
P3	Fleet analytics / historical dispatch logs lower or under Overview.

Dispatch

[Overview] [Drivers] [Verification]

ACTIVE LOADS (assign now)

• Load #88 OPEN [pick a driver ▼] [Assign]

• Load #90 OFFERED ⌚ 04:12

tab content (Overview KPIs, etc.)

P1 (was tab #4)

P2/P3

4.3 Owner-Operator Dashboard

Tier	Content (all existing)
P1	"Active Load Offers" StatCard + Available Loads list with accept action (already near top — keep, add a countdown like Driver).
P2	Your Fleet card with an inline add/assign-driver action (D8), plus the link to full fleet settings.
P3	Analytics + factoring entry points (already routed) below.

4.4 Loadboard (shared list pattern — Shipper "All loads", Carrier "Dispatch", OO "Available")

- **Sort/priority:** time-sensitive first — OFFERED /expiring (with a live Countdown), then OPEN , then in-transit; terminal states (DELIVERED / CANCELLED) collapse to a "History" section (P3).
- **Row density:** show the 4–5 decision columns (lane, rate, equipment, status, action); push the rest into a row-expander so the grid stops horizontally scrolling on ≤1080p (fixes D5 without hiding data).
- **Empty/urgent states:** a clear "No loads need action" vs. a highlighted "3 loads awaiting your response."

4.5 Settings (all personas)

Regroup the 7 tabs (no field removed): **Profile · Equipment** → *Operations*; **ID + Business Verification** → one **Verification** tab with a completion badge (fixes D3); **Authority & Insurance · Organisation** → *Business*; **Security** stays. Deep-links (? tab=...) preserved.

5. Recommendations & Sequencing (post-approval)

Priority	Item	Findings closed	Effort
1	Fix the fake Shipper chart (wire real data or relabel)	D1	S
2	Promote Carrier Dispatch + Shipper "Needs attention" above the fold	D2, D9	M
3	Merge Verification into one tab + add the disabled-CTA hint	D3, D4	S
4	Responsive table strategy (kill horizontal scroll)	D5, D11	M
5	Route-level code-splitting	D6	M
6	Playwright cross-persona nav + 0-console-error crawl (QA automation)	D12, D10	M
7	Settings regroup + OO inline fleet action	D7, D8	M

All changes are additive/organizational. No feature, filter, button, or data point is deleted — the audit's mandate is visibility and flow.

Prepared for review. No implementation, layout change, or live-environment action will be taken until this report is explicitly approved.

Round 2 — Rendered-UI Audit (screenshot evidence)

Method: Round 1 was code-derived. Round 2 reviews **live rendered screenshots** (Owner-Operator dashboard x2, Shipper dashboard, Post-a-load, Load History) and traces each visual issue back to source. Rendered evidence surfaced a **structural composition bug** and a set of **visual-finish** issues that static analysis could not.

R2.0 Headline (the three stakeholder callouts — all confirmed)

1. **"Settings appears twice" — CONFIRMED (High).** On the OO dashboard, *Settings* is in the left sidebar (Account group) **and** repeated as a button in the page header (`OwnerOperatorDashboard.tsx:79–80`). Two controls, same destination, ~6 cm apart.
2. **"Live map + load are below the fold" — CONFIRMED (High).** The OO's only actionable content — the **Available Loads** row (`Dallas → Houston, 1 offered`) and the **Route preview** map — render at the very **bottom** of the page, beneath My haul, verification, an all-zero "Alerts" wall, Fleet & compliance, and a "Tendered loadboard". A driver-operator must scroll past ~6 mostly-empty sections to reach the load they can actually accept.
3. **"White cards look unfinished / like paper on the background" — CONFIRMED (Med).** White surfaces sit on a saturated blue→violet page gradient with inconsistent elevation and large empty gradient voids (Load History, Shipper dashboard tail). The result reads as unfinished "paper floating on a backdrop."

R2.1 Root cause of #1 and #2 — a stacked-dashboard composition bug

`pages/owner-operator/OwnerOperatorDashboard.tsx` renders the **entire** blended dashboard **twice over**:

- `:90` → `<OwnerOperatorDashboardView />` — a complete dashboard (My haul → verification → Alerts → Fleet & compliance → **Tendered loadboard** → Financial → SLA).
- `:94–104` → a **second** set of `StatCard`s (Active Load Offers / Fleet Drivers / MC Number).
- `:118` → a **second** loadboard ("**Available Loads**").
- `:246` → the **Route preview** map.
- then "Your Fleet".

So the page is *[full dashboard] + [another metrics+loadboard+map shell]*. This one bug produces **three** of the visible problems: the map/loads land last (callout #2), there are **two conflicting loadboards** (the View's "Tendered loadboard" shows *"No outstanding offers"* while the shell's "Available Loads" shows *"1 offered"* — directly contradictory), and metrics are duplicated. **Fix is consolidation, not deletion:** render one dashboard; lift the Route map + Available Loads into the top (P1) of the View; collapse the two loadboards into one source of truth.

R2.2 Round-2 Defect Log

#	Severity	Page / Component	Rendered defect (evidence)	Suggested fix (non-destructive)
V1	High	OO Dashboard (<code>OwnerOperatorDashboard.tsx:79–80</code>)	Settings rendered twice — sidebar + header button, same target.	Remove the header duplicate (keep the sidebar canonical), or repurpose the header slot for a real action (e.g. "Go online").
V2	High	OO Dashboard (<code>:90</code> vs <code>:94–246</code>)	Map + Available Loads below the fold ; page stacks two dashboards → the only actionable load & the route map are dead-last.	Lift Route map + Available Loads to P1 ; render a single dashboard (R2.1).
V3	High	OO Dashboard	Two contradictory loadboards — "Tendered loadboard: No outstanding offers" vs "Available Loads: 1 offered" on the same screen.	One loadboard, one truth; dedupe the shell vs View.

#	Severity	Page / Component	Rendered defect (evidence)	Suggested fix (non-destructive)
V4	Med	Global theme	"Paper on background" — white cards on a blue/violet gradient, inconsistent shadows/borders, large empty gradient voids.	Commit to one surface system: either a calm neutral page bg with clearly-elevated cards (border + soft shadow), or tint the cards to sit <i>in</i> the gradient. Constrain max-content width so cards don't float in a void.
V5	Med	OO Dashboard — "Alerts" section	Section titled "Alerts" but every tile reads 0 / "Connect ELD" / "Connect reefer telemetry" / "Pending data" — a wall of empty above real content, and "Alerts" is a misnomer when there are none.	Collapse empty/again-later tiles into a compact "Fleet health" strip; only expand when non-zero; move below the loadboard. Rename to reflect state.
V6	Med	Post a load (RouteMapCard.tsx map embed)	With no address entered, the Route preview shows the entire world map (zoomed all the way out) — reads as broken.	Show a neutral placeholder ("Enter pickup & delivery to preview route") until both ends resolve; don't render a world view.
V7	Med	Shipper Dashboard (ShipperDashboard.tsx:144–152)	The "Match velocity (7 days)" chart renders empty (no bars, just M–S axis) yet still states "Average match: 52s" — reinforces R1-D1: fabricated + visually broken.	Same as D1 — wire real data or hide until data exists; never show a stat with an empty chart.
V8	Low	Top nav / tour layer	Clipped text bleeding over the top edge — fragments "...out this field", "leads." overlap the viewport top/logo (tour tooltip or logo tagline overflow).	Constrain tour tooltip within viewport; fix logo tagline wrap/clip at the top bar.
V9	Low	Load History + Shipper tail	Large empty gradient voids — a single small card floats in a tall purple expanse (short content, full-height gradient).	Center/constrain content, add an illustrative empty state, or cap page height so the void doesn't dominate.

R2.3 Revised Owner-Operator hierarchy (addresses callouts #1 & #2)

Welcome back, Demo Owner Operator LLC [Go online ▶] [Dispatcher Exec] ↳ Refresh	header: NO 2nd Settings (V1)
① My haul: no active load AVAILABLE LOADS (1) • Dallas→Houston 34,000lb \$— [View] [Route] ROUTE PREVIEW Dallas→Houston [map]	P1 P1 ← lifted from bottom (V2)
My verification ✓ ✓ (both gates) Your fleet (1 self · 0 fleet) [+ inline add driver]	P2 P2
▶ Fleet health (ELD/reefer/HOS) ▶ Financial ▶ SLA	P3 (was the "Alerts" zero-wall, V5)

One loadboard (V3). Map + loads above the fold (V2). Settings only in the sidebar (V1). Empty telemetry demoted (V5). **No feature removed** — every element is retained, only relocated/merged.

R2.4 Visual-finish direction (callout #3)

The gradient-with-floating-white-paper look is the single biggest "unfinished" signal. Recommended, low-risk direction (design-token level, applied globally):

- **Surface contrast:** give every card a consistent `1px` hairline border **and** a single soft shadow token, so white reads as a deliberate raised surface, not a cut-out.
- **Background:** reduce gradient saturation (or confine it to the header band) so the body is a calm neutral; the current full-page violet is what makes white cards look pasted-on.
- **Density / voids:** cap main content width (~1200px) and add real empty-states so short pages (Load History) don't leave a tall colored void.
- **Consistency:** audit `bg-card` vs `bg-white` usage so all cards share one treatment.

R2.5 Updated sequencing (Round 1 + Round 2)

Pri	Item	Closes	Effort
1	OO dashboard: de-stack the double render; lift map+loads to P1; one loadboard; drop 2nd Settings	V1,V2,V3, R1-D8	M
2	Visual-finish pass: card border/shadow token, calmer bg, max-width, empty states	V4,V9, callout #3	M
3	Kill fabricated/empty analytics (Shipper chart) + empty-map placeholder	V7,V6, R1-D1	S
4	Demote empty telemetry "Alerts" wall; rename to state	V5	S
5	Carrier Dispatch above the fold + Shipper "needs attention" (R1)	R1-D2,D9	M
6	Settings verification merge + disabled-CTA hint (R1)	R1-D3,D4	S
7	Responsive tables, code-splitting, tour-clip fix, Playwright crawl (R1)	R1-D5,D6,V8,D12	M

Round 2 remains non-destructive and advisory. No code, layout, or live-environment change will be made until this report is explicitly approved.